

GRAPH SKETCHING MODULE COMPLETE SOLUTIONS

MATHEMATICS QA016

Instructions:

- 1) Sketch each function on the axes provided.
- 2) From your graph, determine the corresponding domain and range.

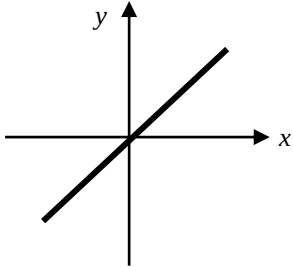
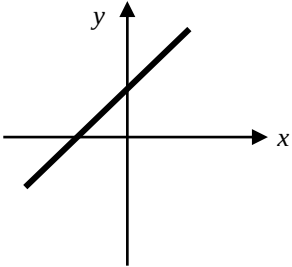
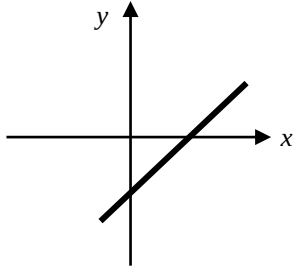
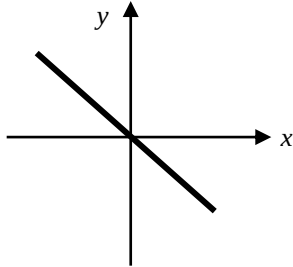
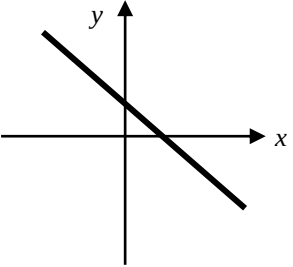
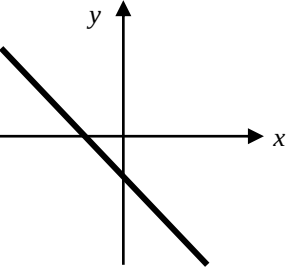
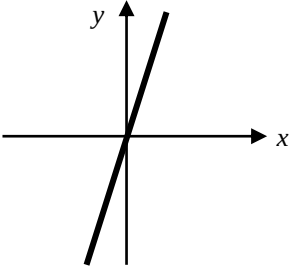
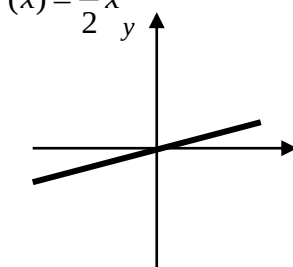
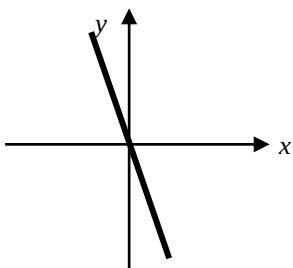
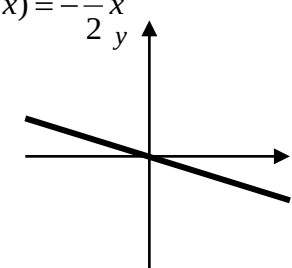
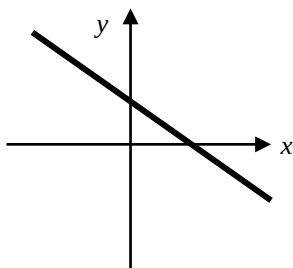
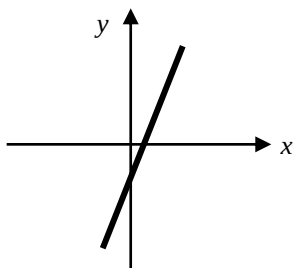
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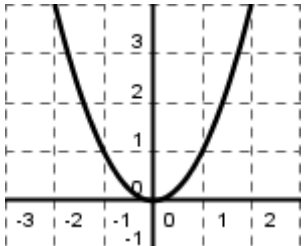
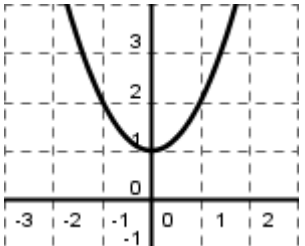
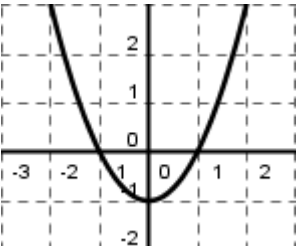
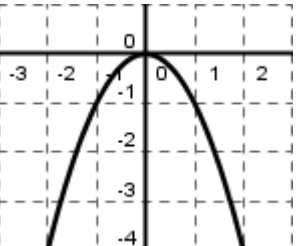
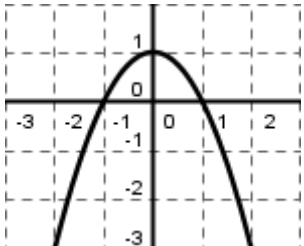
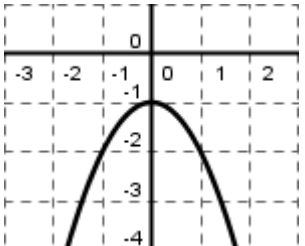
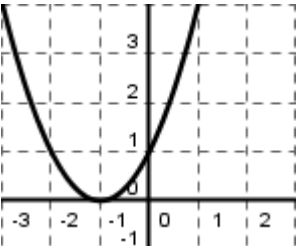
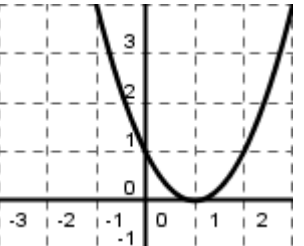
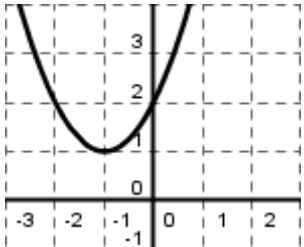
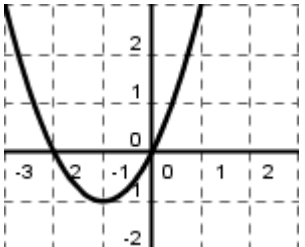
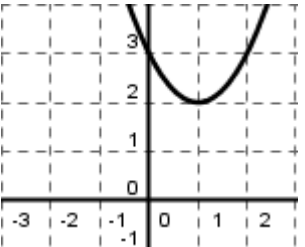
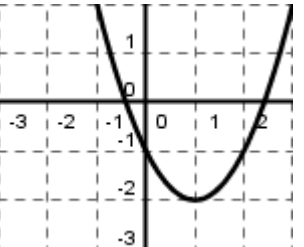
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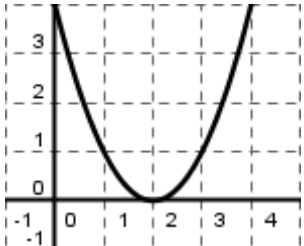
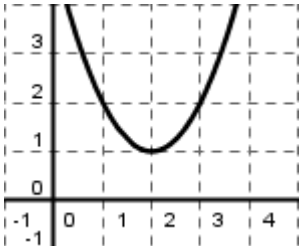
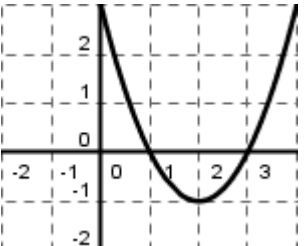
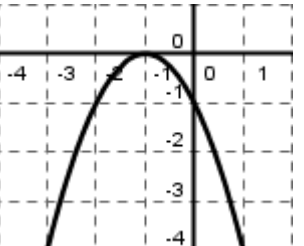
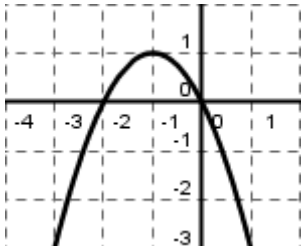
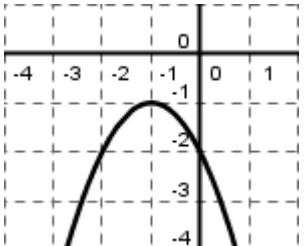
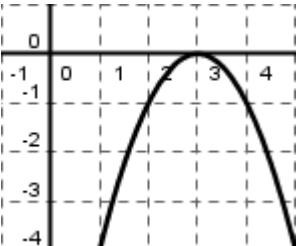
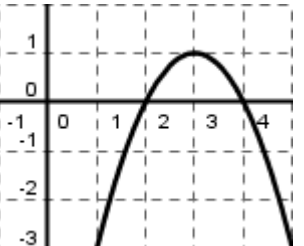
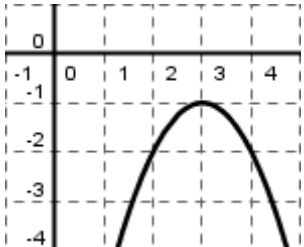
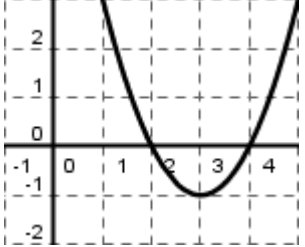
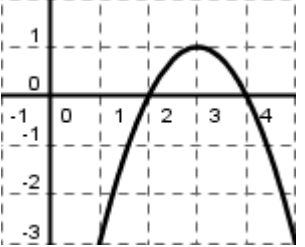
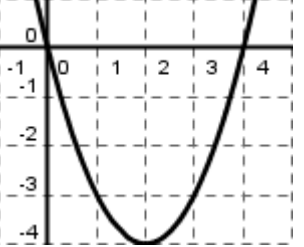
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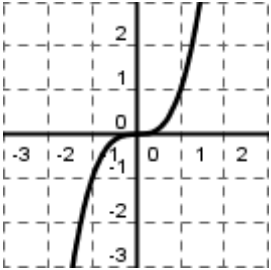
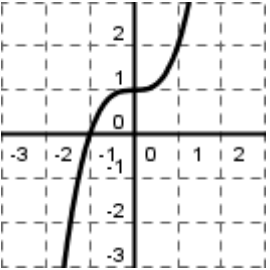
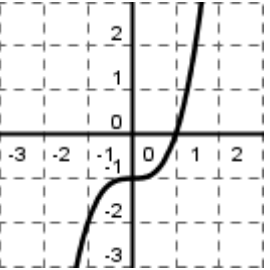
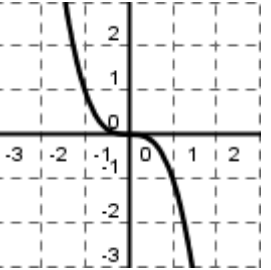
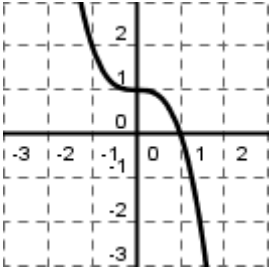
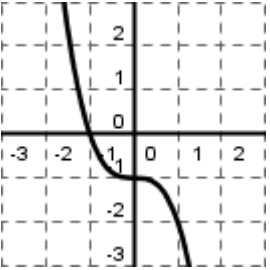
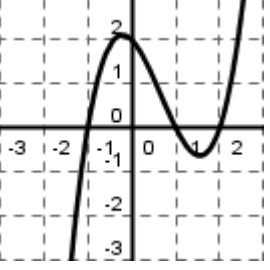
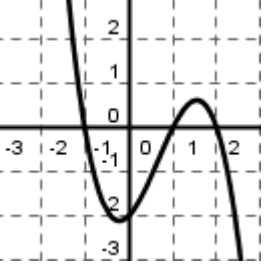
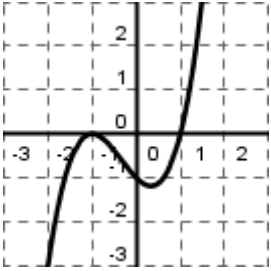
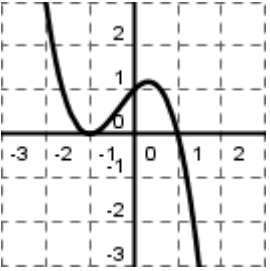
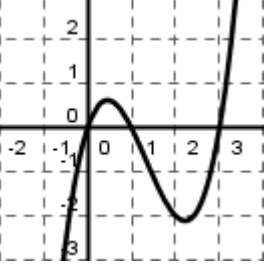
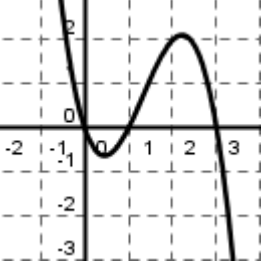
Page 2	: Linear functions
Page 3 - 4	: Quadratic functions
Page 5	: Cubic functions
Page 6	: Absolute value functions
Page 7 - 9	: Square root functions
Page 10 - 14	: Exponential functions
Page 15 - 16	: Logarithmic functions

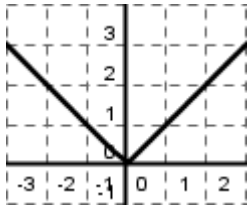
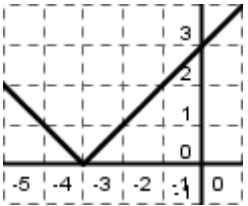
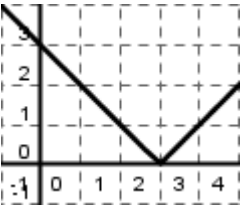
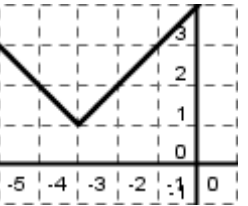
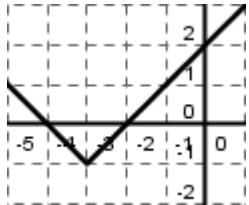
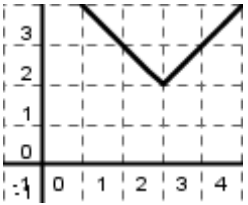
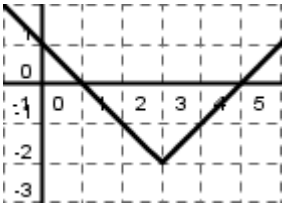
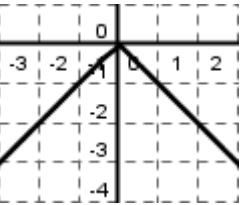
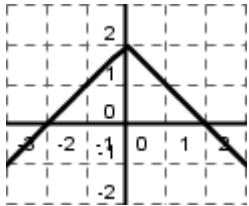
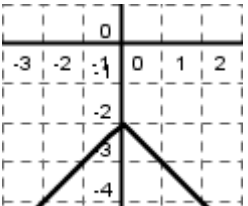
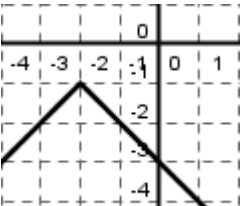
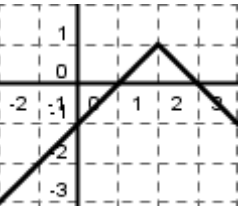
Prepared by Mr Muhammad Fazdhly bin Abdul Muttalib (KMS 2014)
p/s: if there are any mistakes in the answer, please correct them by yourself. ☺

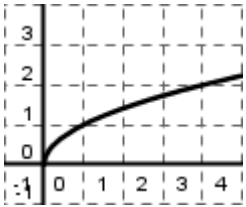
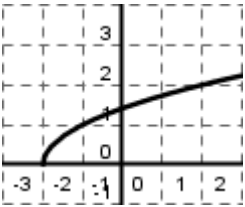
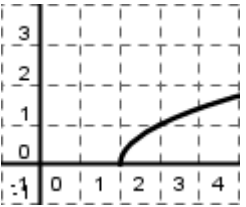
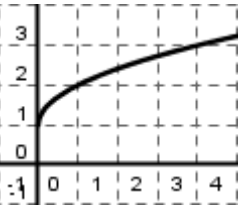
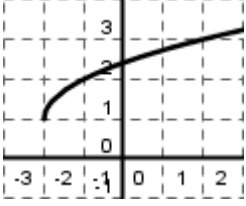
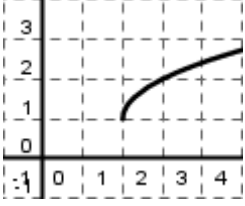
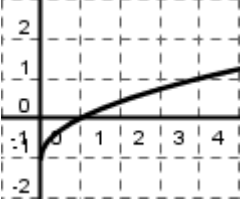
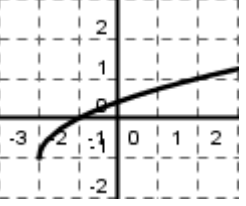
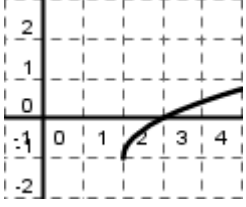
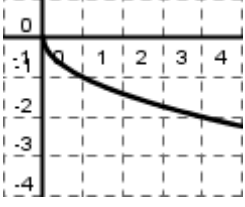
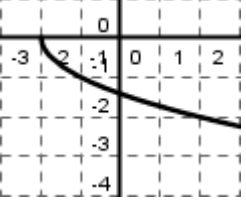
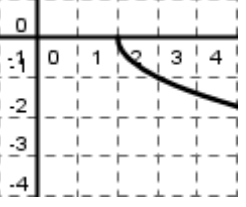
1. $f(x) = x, x \in \mathbb{R}$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$	2. $f(x) = x + 2$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$	3. $f(x) = x - 2$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$	4. $f(x) = -x$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$
5. $f(x) = -x + 2$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$	6. $f(x) = -x - 2$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$	7. $f(x) = 2x$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$	8. $f(x) = \frac{1}{2}x$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$
9. $f(x) = -2x$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$	10. $f(x) = -\frac{1}{2}x$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$	11. $f(x) = 3 - x$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$	12. $f(x) = 2x - 1$  $D_f = (-\infty, \infty)$ $R_f = (-\infty, \infty)$

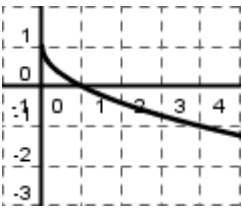
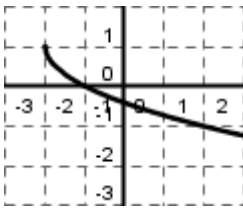
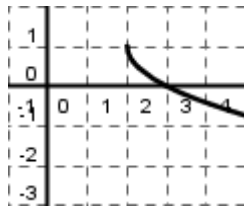
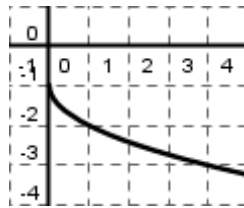
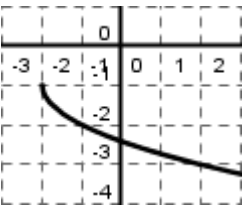
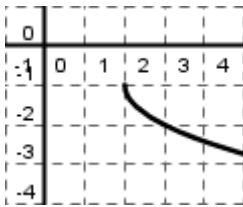
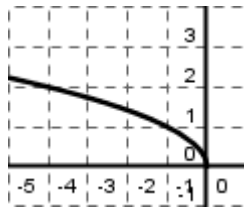
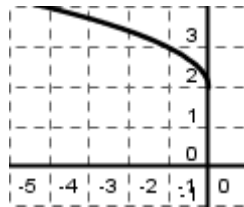
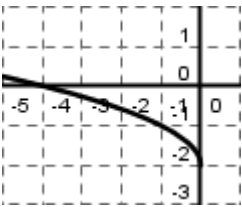
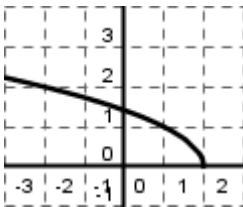
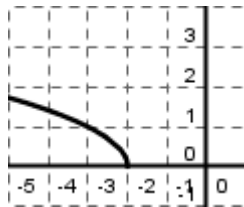
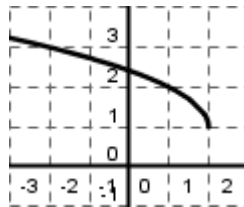
13. $f(x) = x^2, x \in \mathbb{R}$  $D_f = (-\infty, \infty)$ $R_f = 0, \infty$	14. $f(x) = x^2 + 1$  $D_f = (-\infty, \infty)$ $R_f = 1, \infty$	15. $f(x) = x^2 - 1$  $D_f = (-\infty, \infty)$ $R_f = -1, \infty$	16. $f(x) = -x^2$  $D_f = (-\infty, \infty)$ $R_f = -\infty, 0$
17. $f(x) = -x^2 + 1$  $D_f = (-\infty, \infty)$ $R_f = -\infty, 1$	18. $f(x) = -x^2 - 1$  $D_f = (-\infty, \infty)$ $R_f = -\infty, -1$	19. $f(x) = (x+1)^2$  $D_f = (-\infty, \infty)$ $R_f = 0, \infty$	20. $f(x) = (x-1)^2$  $D_f = (-\infty, \infty)$ $R_f = 0, \infty$
21. $f(x) = (x+1)^2 + 1$  $D_f = (-\infty, \infty)$ $R_f = 1, \infty$	22. $f(x) = (x+1)^2 - 1$  $D_f = (-\infty, \infty)$ $R_f = -1, \infty$	23. $f(x) = (x-1)^2 + 2$  $D_f = (-\infty, \infty)$ $R_f = 2, \infty$	24. $f(x) = (x-1)^2 - 2$  $D_f = (-\infty, \infty)$ $R_f = -2, \infty$

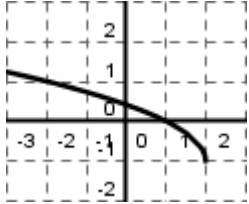
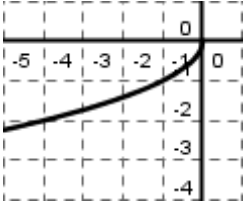
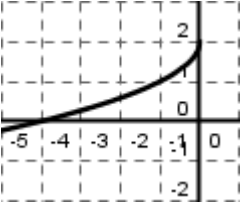
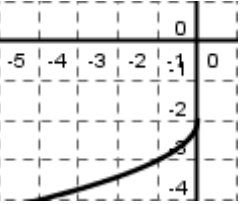
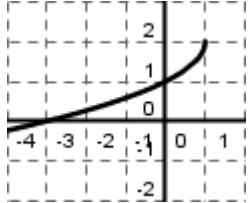
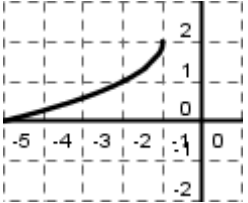
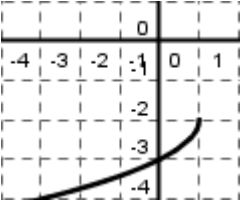
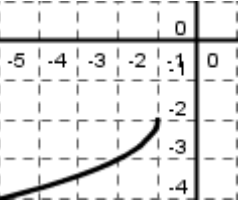
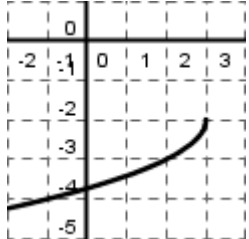
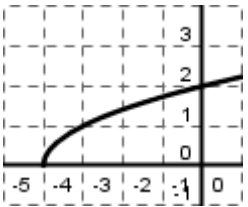
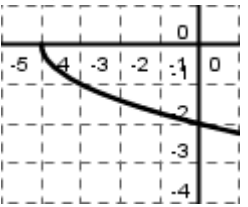
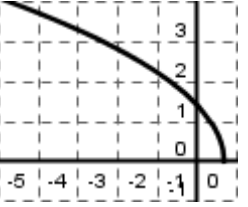
25. $f(x) = (2-x)^2$  $D_f = -\infty, \infty$ $R_f = 0, \infty$	26. $f(x) = (2-x)^2 + 1$  $D_f = -\infty, \infty$ $R_f = 1, \infty$	27. $f(x) = (2-x)^2 - 1$  $D_f = -\infty, \infty$ $R_f = -1, \infty$	28. $f(x) = -(x+1)^2$  $D_f = -\infty, \infty$ $R_f = -\infty, 0$
29. $f(x) = -(x+1)^2 + 1$  $D_f = -\infty, \infty$ $R_f = -\infty, 1$	30. $f(x) = -(x+1)^2 - 1$  $D_f = -\infty, \infty$ $R_f = -\infty, -1$	31. $f(x) = -(x-3)^2$  $D_f = -\infty, \infty$ $R_f = -\infty, 0$	32. $f(x) = -(x-3)^2 + 1$  $D_f = -\infty, \infty$ $R_f = -\infty, 1$
33. $f(x) = -(x-3)^2 - 1$  $D_f = -\infty, \infty$ $R_f = -\infty, -1$	34. $f(x) = (x-4)(x-2)$  $D_f = -\infty, \infty$ $R_f = -1, \infty$	35. $f(x) = -(x-4)(x-2)$  $D_f = -\infty, \infty$ $R_f = -\infty, 1$	36. $f(x) = x(x-4)$  $D_f = -\infty, \infty$ $R_f = -4, \infty$

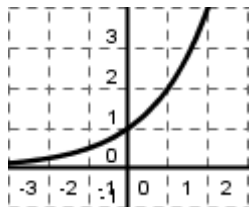
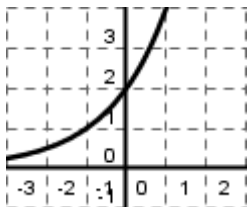
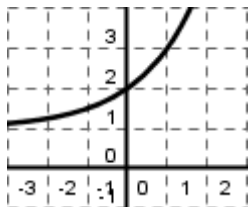
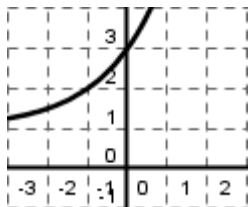
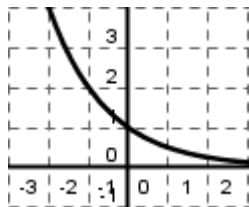
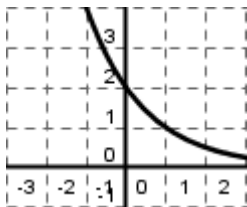
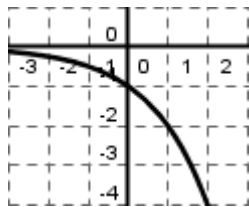
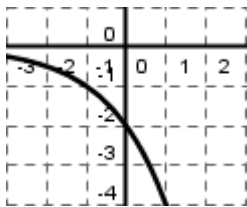
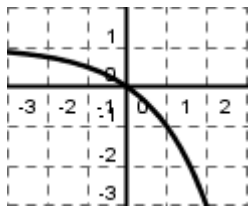
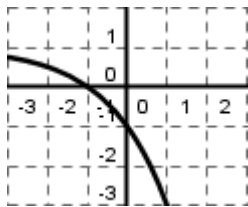
37. $f(x) = x^3$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$	38. $f(x) = x^3 + 1$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$	39. $f(x) = x^3 - 1$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$	40. $f(x) = -x^3$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$
41. $f(x) = -x^3 + 1$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$	42. $f(x) = -x^3 - 1$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$	43. $f(x) = (x-2)(x-1)(x+1)$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$	44. $f(x) = -(x-2)(x-1)(x+1)$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$
45. $f(x) = (x-1)(x+1)^2$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$	46. $f(x) = -(x-1)(x+1)^2$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$	47. $f(x) = x(x-1)(x-3)$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$	48. $f(x) = -x(x-1)(x-3)$  $D_f = -\infty, \infty$ $R_f = -\infty, \infty$

49. $f(x) = x , x \in \mathbb{R}$  $D_f = -\infty, \infty$ $R_f = 0, \infty$	50. $f(x) = x+3$  $D_f = -\infty, \infty$ $R_f = 0, \infty$	51. $f(x) = x-3$  $D_f = -\infty, \infty$ $R_f = 0, \infty$	52. $f(x) = x+3 +1$  $D_f = -\infty, \infty$ $R_f = 1, \infty$
53. $f(x) = x+3 -1$  $D_f = -\infty, \infty$ $R_f = -1, \infty$	54. $f(x) = x-3 +2$  $D_f = -\infty, \infty$ $R_f = 2, \infty$	55. $f(x) = x-3 -2$  $D_f = -\infty, \infty$ $R_f = -2, \infty$	56. $f(x) = - x$  $D_f = -\infty, \infty$ $R_f = -\infty, 0$
57. $f(x) = - x +2$  $D_f = -\infty, \infty$ $R_f = -\infty, 2$	58. $f(x) = - x -2$  $D_f = -\infty, \infty$ $R_f = -\infty, -2$	59. $f(x) = - x+2 -1$  $D_f = -\infty, \infty$ $R_f = -\infty, -1$	60. $f(x) = - x-2 +1$  $D_f = -\infty, \infty$ $R_f = -\infty, 1$

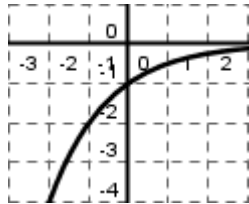
61. $f(x) = \sqrt{x}$  $D_f = 0, \infty$ $R_f = 0, \infty$	62. $f(x) = \sqrt{x+2}$  $D_f = -2, \infty$ $R_f = 0, \infty$	63. $f(x) = \sqrt{x-2}$  $D_f = 2, \infty$ $R_f = 0, \infty$	64. $f(x) = \sqrt{x} + 1$  $D_f = 0, \infty$ $R_f = 1, \infty$
65. $f(x) = \sqrt{x+2} + 1$  $D_f = -2, \infty$ $R_f = 1, \infty$	66. $f(x) = \sqrt{x-2} + 1$  $D_f = 2, \infty$ $R_f = 1, \infty$	67. $f(x) = \sqrt{x} - 1$  $D_f = 0, \infty$ $R_f = -1, \infty$	68. $f(x) = \sqrt{x+2} - 1$  $D_f = -2, \infty$ $R_f = -1, \infty$
69. $f(x) = \sqrt{x-2} - 1$  $D_f = 2, \infty$ $R_f = -1, \infty$	70. $f(x) = -\sqrt{x}$  $D_f = 0, \infty$ $R_f = -\infty, 0$	71. $f(x) = -\sqrt{x+2}$  $D_f = -2, \infty$ $R_f = -\infty, 0$	72. $f(x) = -\sqrt{x-2}$  $D_f = 2, \infty$ $R_f = -\infty, 0$

73. $f(x) = -\sqrt{x} + 1$  $D_f = 0, \infty$ $R_f = -\infty, 1$	74. $f(x) = -\sqrt{x+2} + 1$  $D_f = -2, \infty$ $R_f = -\infty, 1$	75. $f(x) = -\sqrt{x-2} + 1$  $D_f = 2, \infty$ $R_f = -\infty, 1$	76. $f(x) = -\sqrt{x} - 1$  $D_f = 0, \infty$ $R_f = -\infty, -1$
77. $f(x) = -\sqrt{x+2} - 1$  $D_f = -2, \infty$ $R_f = -\infty, -1$	78. $f(x) = -\sqrt{x-2} - 1$  $D_f = 2, \infty$ $R_f = -\infty, -1$	79. $f(x) = \sqrt{-x}$  $D_f = -\infty, 0$ $R_f = 0, \infty$	80. $f(x) = \sqrt{-x} + 2$  $D_f = -\infty, 0$ $R_f = 2, \infty$
81. $f(x) = \sqrt{-x} - 2$  $D_f = -\infty, 0$ $R_f = -2, \infty$	82. $f(x) = \sqrt{-x+2}$  $D_f = -\infty, 2$ $R_f = 0, \infty$	83. $f(x) = \sqrt{-x-2}$  $D_f = -\infty, -2$ $R_f = 0, \infty$	84. $f(x) = \sqrt{-x+2} + 1$  $D_f = -\infty, 2$ $R_f = 1, \infty$

85. $f(x) = \sqrt{-x+2} - 1$  $D_f = -\infty, 2$ $R_f = -1, \infty$	86. $f(x) = -\sqrt{-x}$  $D_f = -\infty, 0$ $R_f = -\infty, 0$	87. $f(x) = -\sqrt{-x} + 2$  $D_f = -\infty, 0$ $R_f = -\infty, 2$	88. $f(x) = -\sqrt{-x} - 2$  $D_f = -\infty, 0$ $R_f = -\infty, -2$
89. $f(x) = -\sqrt{-x+1} + 2$  $D_f = -\infty, 1$ $R_f = -\infty, 2$	90. $f(x) = -\sqrt{-x-1} + 2$  $D_f = -\infty, -1$ $R_f = -\infty, 2$	91. $f(x) = -\sqrt{-x+1} - 2$  $D_f = -\infty, 1$ $R_f = -\infty, -2$	92. $f(x) = -\sqrt{-x-1} - 2$  $D_f = -\infty, -1$ $R_f = -\infty, -2$
93. $f(x) = -\sqrt{3-x} - 2$  $D_f = -\infty, 3$ $R_f = -\infty, -2$	94. $f(x) = (4+x)^{\frac{1}{2}}$  $D_f = -4, \infty$ $R_f = 0, \infty$	95. $f(x) = -(4+x)^{\frac{1}{2}}$  $D_f = -4, \infty$ $R_f = -\infty, 0$	96. $f(x) = \sqrt{2-3x}$  $D_f = -\infty, 2/3$ $R_f = 0, \infty$

97. $f(x) = 2^x$  $D_f = -\infty, \infty$ $R_f = 0, \infty$	98. $f(x) = 2^{x+1}$  $D_f = -\infty, \infty$ $R_f = 0, \infty$	99. $f(x) = 2^x + 1$  $D_f = -\infty, \infty$ $R_f = 1, \infty$	100. $f(x) = 2^{x+1} + 1$  $D_f = -\infty, \infty$ $R_f = 1, \infty$
101. $f(x) = 2^{-x}$  $D_f = -\infty, \infty$ $R_f = 0, \infty$	102. $f(x) = 2^{-x+1}$  $D_f = -\infty, \infty$ $R_f = 0, \infty$	103. $f(x) = 2^{-x} + 1$  $D_f = -\infty, \infty$ $R_f = 1, \infty$	104. $f(x) = 2^{-x+1} + 1$  $D_f = -\infty, \infty$ $R_f = 1, \infty$
105. $f(x) = -2^x$  $D_f = -\infty, \infty$ $R_f = -\infty, 0$	106. $f(x) = -2^{x+1}$  $D_f = -\infty, \infty$ $R_f = -\infty, 0$	107. $f(x) = -2^x + 1$  $D_f = -\infty, \infty$ $R_f = -\infty, 1$	108. $f(x) = -2^{x+1} + 1$  $D_f = -\infty, \infty$ $R_f = -\infty, 1$

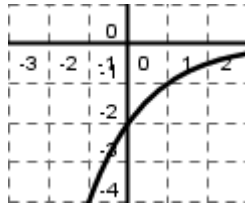
109. $f(x) = -2^{-x}$



$D_f = -\infty, \infty$

$R_f = -\infty, 0$

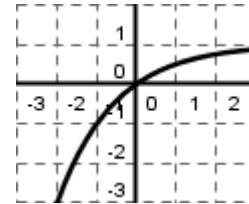
110. $f(x) = -2^{-x+1}$



$D_f = -\infty, \infty$

$R_f = -\infty, 0$

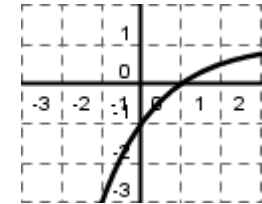
111. $f(x) = -2^{-x} + 1$



$D_f = -\infty, \infty$

$R_f = -\infty, 1$

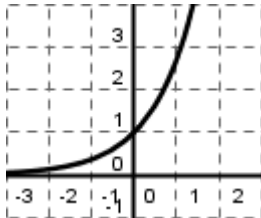
112. $f(x) = -2^{-x+1} + 1$



$D_f = -\infty, \infty$

$R_f = -\infty, 1$

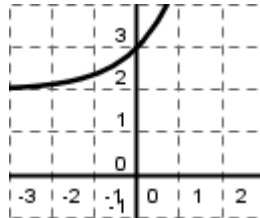
113. $f(x) = e^x$



$D_f = -\infty, \infty$

$R_f = 0, \infty$

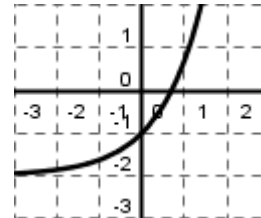
114. $f(x) = e^x + 2$



$D_f = -\infty, \infty$

$R_f = 2, \infty$

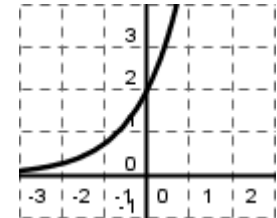
115. $f(x) = e^x - 2$



$D_f = -\infty, \infty$

$R_f = -2, \infty$

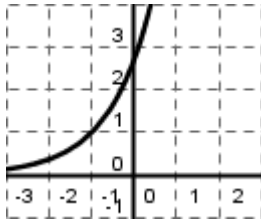
116. $f(x) = 2e^x$



$D_f = -\infty, \infty$

$R_f = 0, \infty$

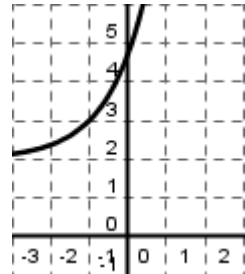
117. $f(x) = e^{x+1}$



$D_f = -\infty, \infty$

$R_f = 0, \infty$

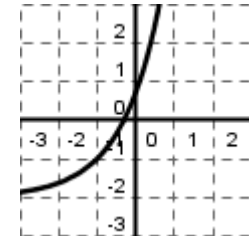
118. $f(x) = e^{x+1} + 2$



$D_f = -\infty, \infty$

$R_f = 2, \infty$

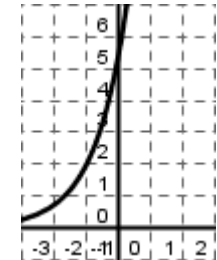
119. $f(x) = e^{x+1} - 2$



$D_f = -\infty, \infty$

$R_f = -2, \infty$

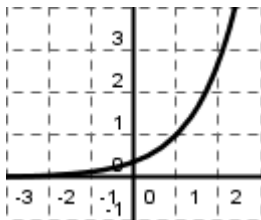
120. $f(x) = 2e^{x+1}$



$D_f = -\infty, \infty$

$R_f = 0, \infty$

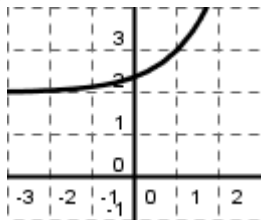
121. $f(x) = e^{x-1}$



$D_f = -\infty, \infty$

$R_f = 0, \infty$

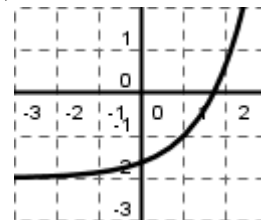
122. $f(x) = e^{x-1} + 2$



$D_f = -\infty, \infty$

$R_f = 2, \infty$

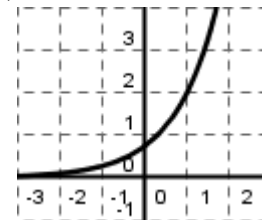
123. $f(x) = e^{x-1} - 2$



$D_f = -\infty, \infty$

$R_f = -2, \infty$

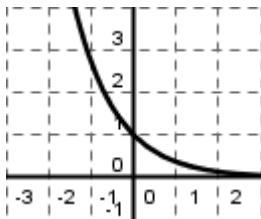
124. $f(x) = 2e^{x-1}$



$D_f = -\infty, \infty$

$R_f = 0, \infty$

125. $f(x) = e^{-x}$



$D_f = -\infty, \infty$

$R_f = 0, \infty$

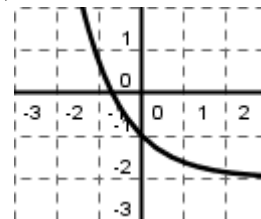
126. $f(x) = e^{-x} + 2$



$D_f = -\infty, \infty$

$R_f = 2, \infty$

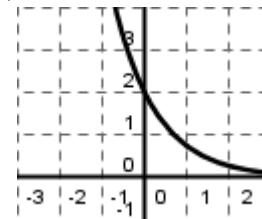
127. $f(x) = e^{-x} - 2$



$D_f = -\infty, \infty$

$R_f = -2, \infty$

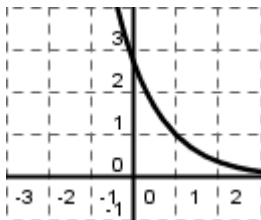
128. $f(x) = 2e^{-x}$



$D_f = -\infty, \infty$

$R_f = 0, \infty$

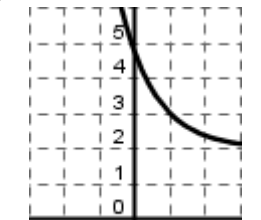
129. $f(x) = e^{-x+1}$



$D_f = -\infty, \infty$

$R_f = 0, \infty$

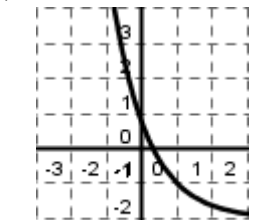
130. $f(x) = e^{-x+1} + 2$



$D_f = -\infty, \infty$

$R_f = 2, \infty$

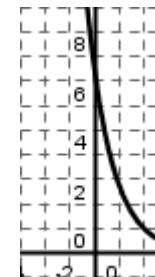
131. $f(x) = e^{-x+1} - 2$



$D_f = -\infty, \infty$

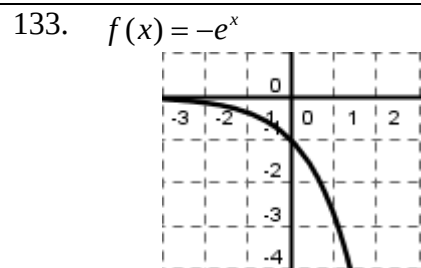
$R_f = -2, \infty$

132. $f(x) = e^{2-x}$



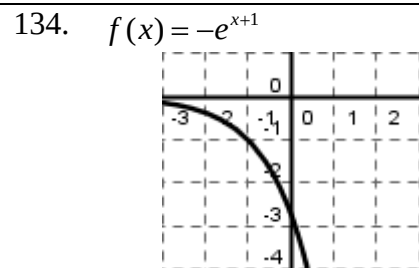
$D_f = -\infty, \infty$

$R_f = 0, \infty$



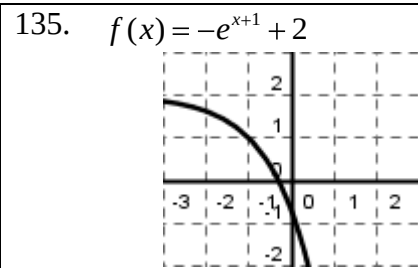
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 0$$



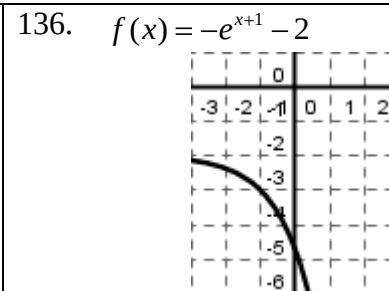
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 0$$



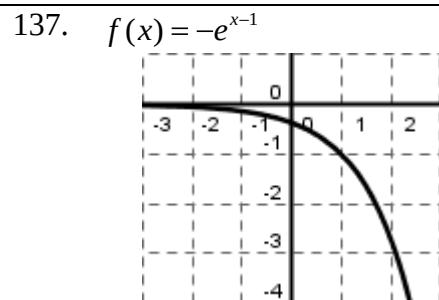
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 2$$



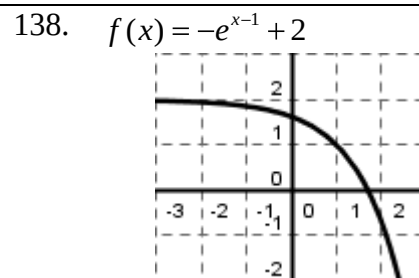
$$D_f = -\infty, \infty$$

$$R_f = -\infty, -2$$



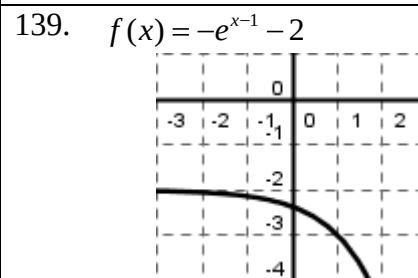
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 0$$



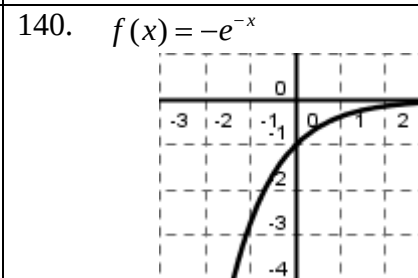
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 2$$



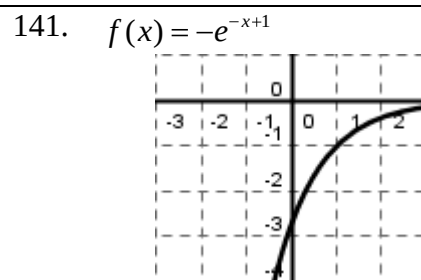
$$D_f = -\infty, \infty$$

$$R_f = -\infty, -2$$



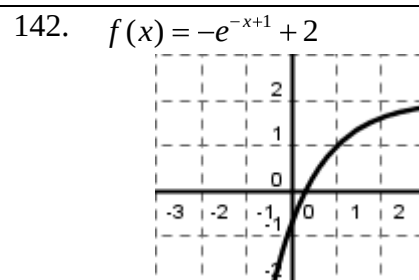
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 0$$



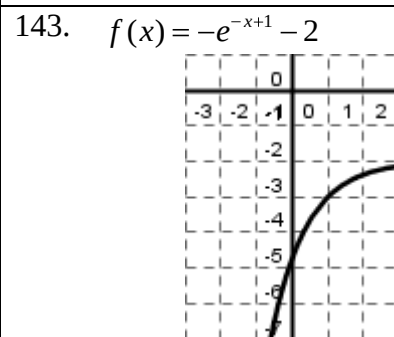
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 0$$



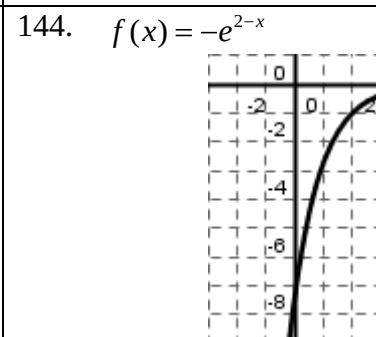
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 2$$



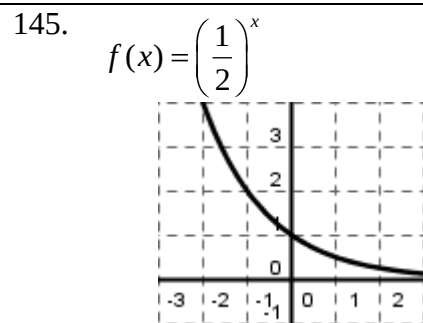
$$D_f = -\infty, \infty$$

$$R_f = -\infty, -2$$



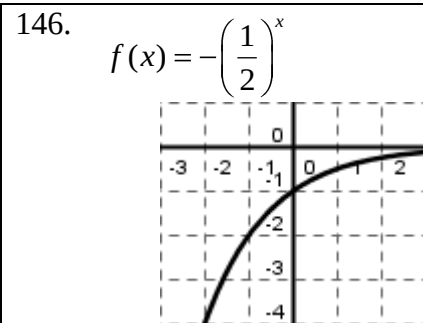
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 0$$



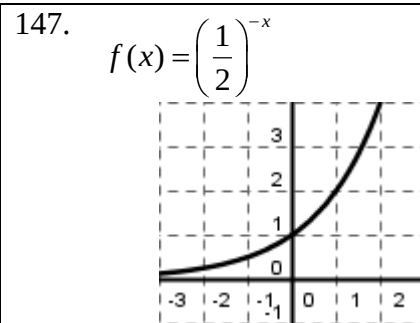
$$D_f = -\infty, \infty$$

$$R_f = 0, \infty$$



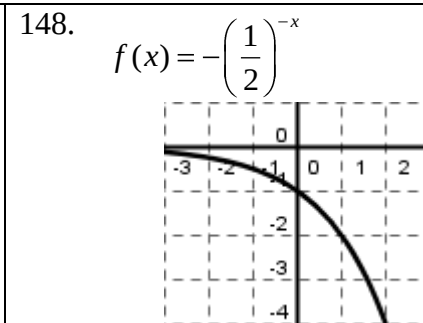
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 0$$



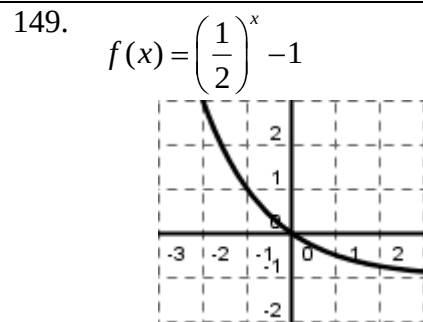
$$D_f = -\infty, \infty$$

$$R_f = 0, \infty$$



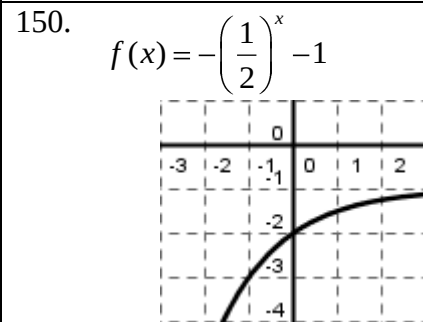
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 0$$



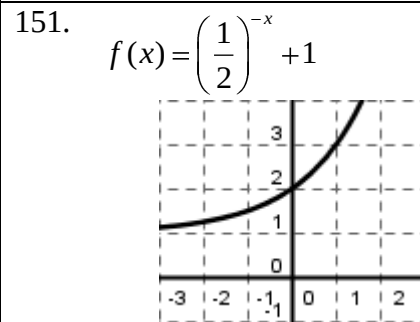
$$D_f = -\infty, \infty$$

$$R_f = -1, \infty$$



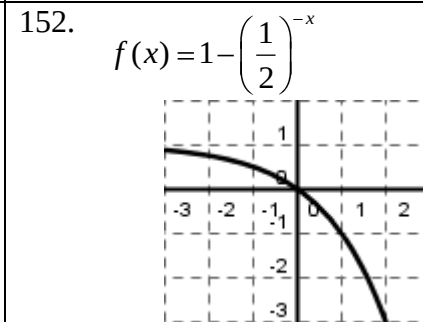
$$D_f = -\infty, \infty$$

$$R_f = -\infty, -1$$



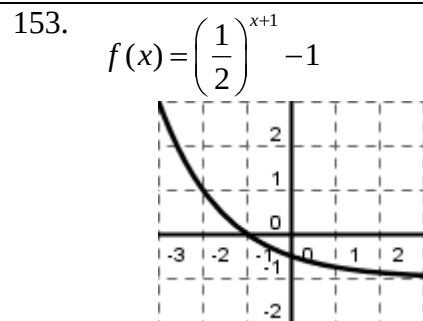
$$D_f = -\infty, \infty$$

$$R_f = 1, \infty$$



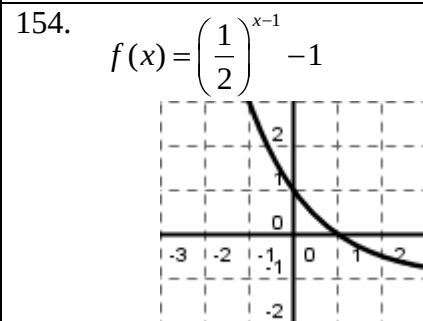
$$D_f = -\infty, \infty$$

$$R_f = -\infty, 1$$



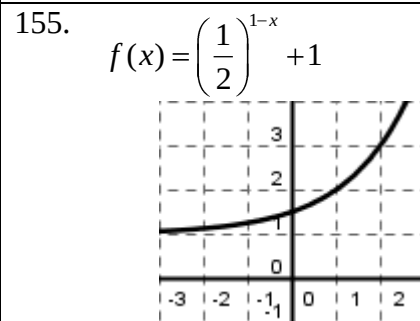
$$D_f = -\infty, \infty$$

$$R_f = -1, \infty$$



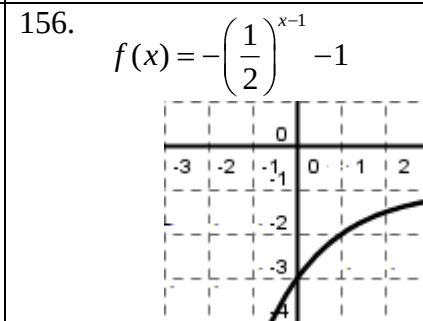
$$D_f = -\infty, \infty$$

$$R_f = -1, \infty$$



$$D_f = -\infty, \infty$$

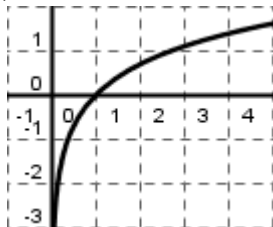
$$R_f = 1, \infty$$



$$D_f = -\infty, \infty$$

$$R_f = -\infty, -1$$

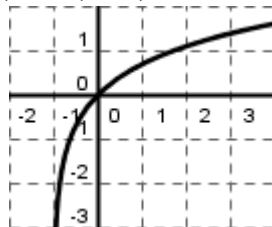
157. $f(x) = \ln x$



$D_f = 0, \infty$

$R_f = -\infty, \infty$

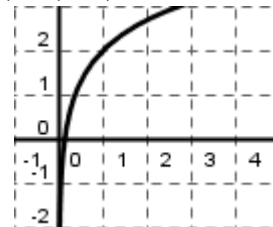
158. $f(x) = \ln(x+1)$



$D_f = -1, \infty$

$R_f = -\infty, \infty$

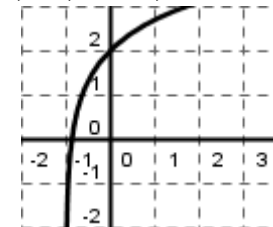
159. $f(x) = (\ln x) + 2$



$D_f = 0, \infty$

$R_f = -\infty, \infty$

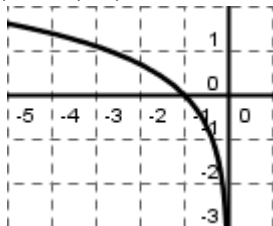
160. $f(x) = (\ln x + 1) + 2$



$D_f = -1, \infty$

$R_f = -\infty, \infty$

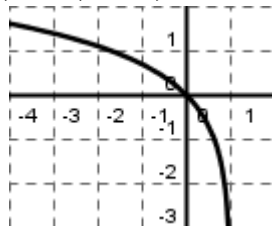
161. $f(x) = \ln(-x)$



$D_f = -\infty, 0$

$R_f = -\infty, \infty$

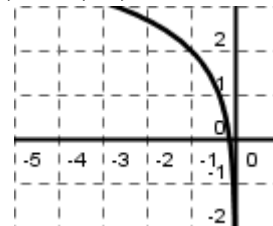
162. $f(x) = \ln(-x+1)$



$D_f = -\infty, 1$

$R_f = -\infty, \infty$

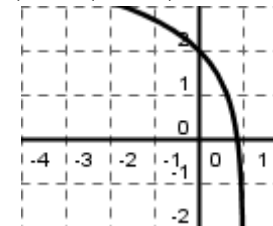
163. $f(x) = \ln(-x) + 2$



$D_f = -\infty, 0$

$R_f = -\infty, \infty$

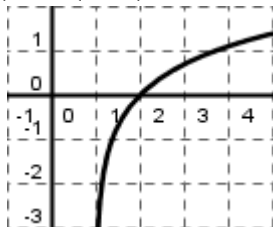
164. $f(x) = \ln(-x+1) + 2$



$D_f = -\infty, 1$

$R_f = -\infty, \infty$

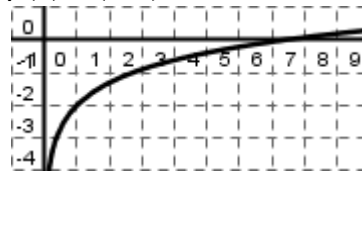
165. $f(x) = \ln(x-1)$



$D_f = 1, \infty$

$R_f = -\infty, \infty$

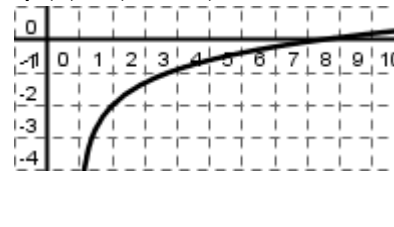
166. $f(x) = (\ln x) - 2$



$D_f = 0, \infty$

$R_f = -\infty, \infty$

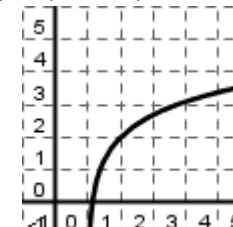
167. $f(x) = (\ln x - 1) - 2$



$D_f = 1, \infty$

$R_f = -\infty, \infty$

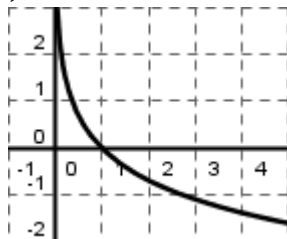
168. $f(x) = (\ln x - 1) + 2$



$D_f = 1, \infty$

$R_f = -\infty, \infty$

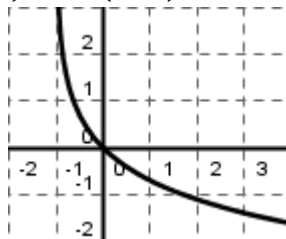
169. $f(x) = -\ln x$



$D_f = 0, \infty$

$R_f = -\infty, \infty$

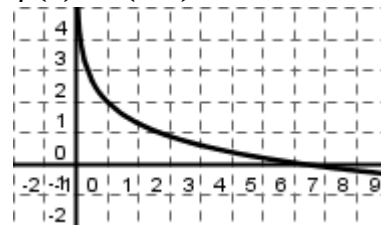
170. $f(x) = -\ln(x+1)$



$D_f = -1, \infty$

$R_f = -\infty, \infty$

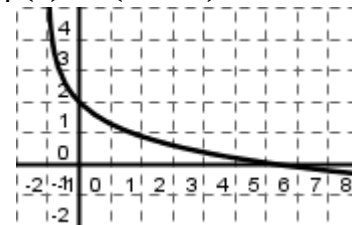
171. $f(x) = -(\ln x) + 2$



$D_f = 0, \infty$

$R_f = -\infty, \infty$

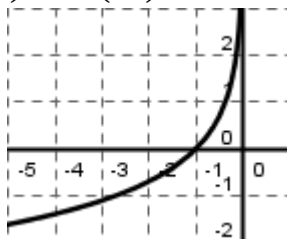
172. $f(x) = -(\ln x + 1) + 2$



$D_f = -1, \infty$

$R_f = -\infty, \infty$

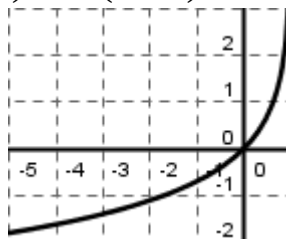
173. $f(x) = -\ln(-x)$



$D_f = -\infty, 0$

$R_f = -\infty, \infty$

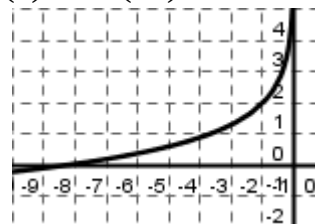
174. $f(x) = -\ln(-x+1)$



$D_f = -\infty, 1$

$R_f = -\infty, \infty$

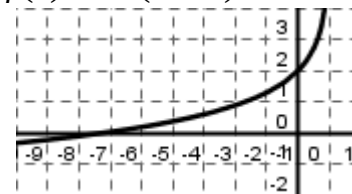
175. $f(x) = -\ln(-x) + 2$



$D_f = -\infty, 0$

$R_f = -\infty, \infty$

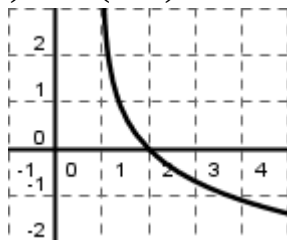
176. $f(x) = -\ln(-x+1) + 2$



$D_f = -\infty, 1$

$R_f = -\infty, \infty$

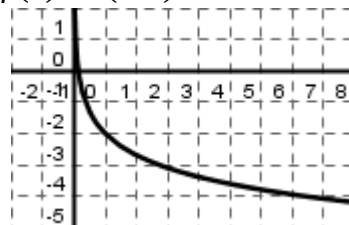
177. $f(x) = -\ln(x-1)$



$D_f = 1, \infty$

$R_f = -\infty, \infty$

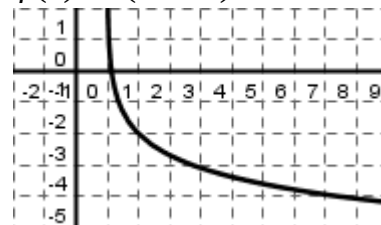
178. $f(x) = -(\ln x) - 2$



$D_f = 0, \infty$

$R_f = -\infty, \infty$

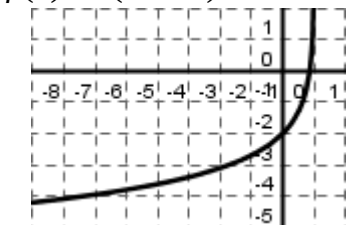
179. $f(x) = -(\ln x - 1) - 2$



$D_f = 1, \infty$

$R_f = -\infty, \infty$

180. $f(x) = -(\ln 1 - x) - 2$



$D_f = -\infty, 1$

$R_f = -\infty, \infty$

